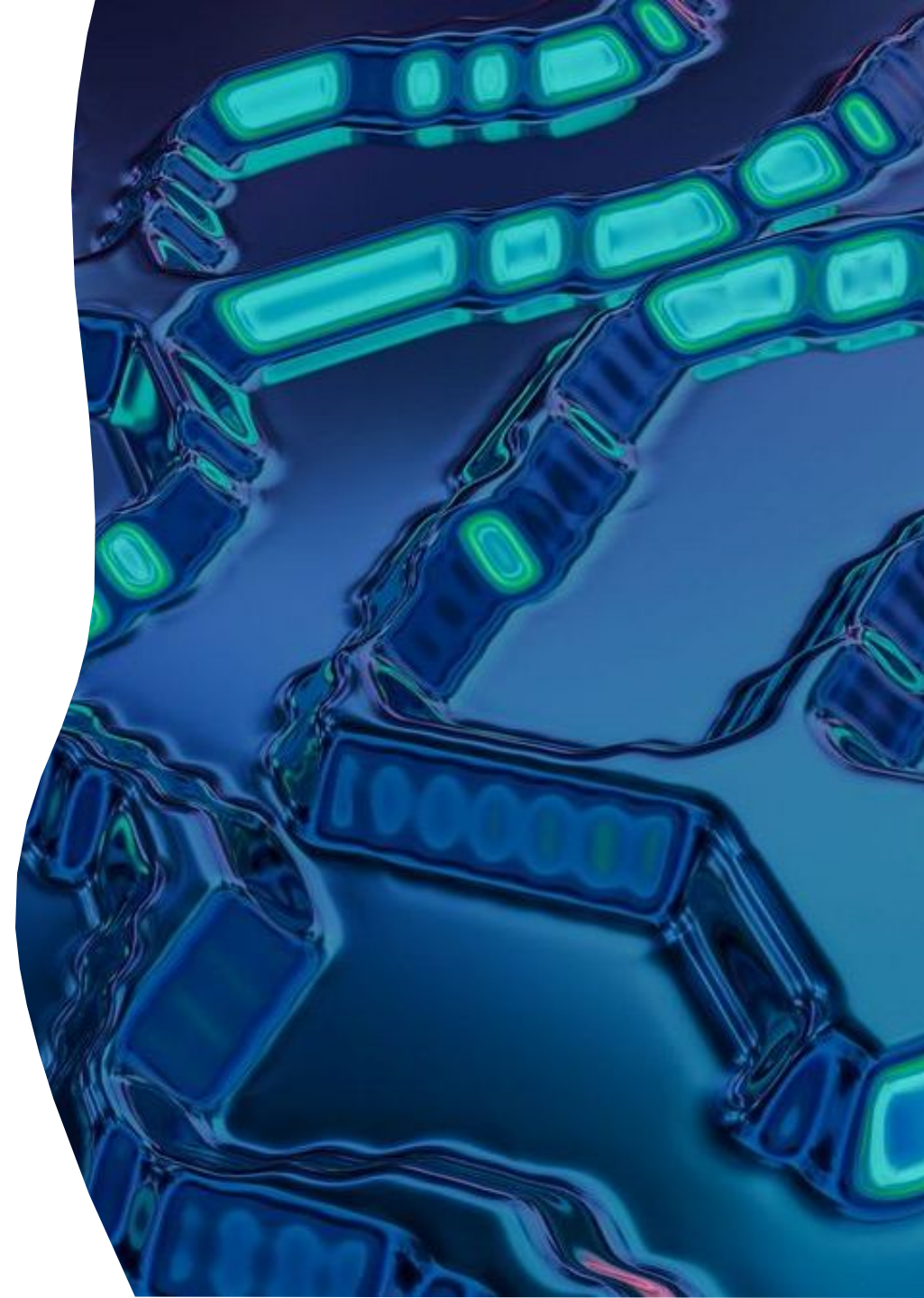


Data Analytics, Data Mining and Machine Learning

Lecture 3

Narges Chinichian · Semester 1, 2026



Overview of this course

Week	Date	Topic
1	07.09.2026	Introduction
2	14.09.2026	Data, Research Questions & Ethics
3	21.09.2026	Data Visualisation & Exploratory Data Analysis
4	28.09.2026	Preprocessing & Feature Engineering
5	05.10.2026	Machine Learning Fundamentals & Regression
6	12.10.2026	Project Presentations I
7	19.10.2026	Classification: Logistic Regression, KNN & Decision Trees
8	26.10.2026	Bank Holiday — No Class
9	02.11.2026	Gradient Descent, Regularisation & Model Complexity
10	09.11.2026	Model Evaluation & Ensemble Methods
11	16.11.2026	Neural Networks, Unsupervised Learning & Special Topics
12	23.11.2026	Project Presentations II

Quick Quiz

Check Loop





Today's Agenda

01

Python refresher

02

Plots Basics

03

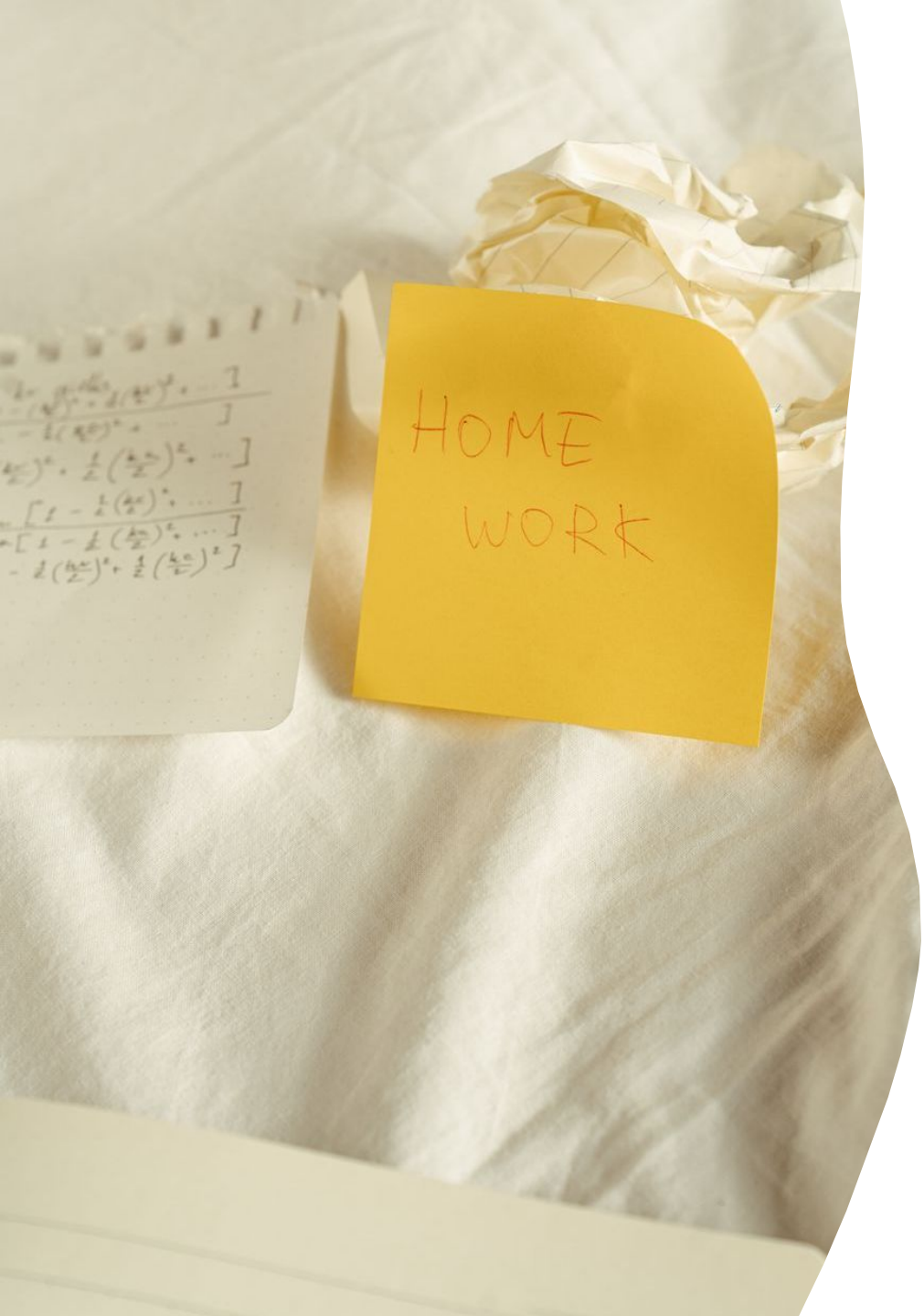
Exploratory Data Analysis (EDA)

Each of these has a notebook on your week's repo. Let's go directly to our notebooks.

The first ten minutes with any dataset (cheatsheet)

The same opening moves every time.

```
df.shape          # how big?
df.head() / df.sample(5)  # what's a row?
df.info()         # types — is a number stored as text?
df.isna().sum()    # what's missing?
df.duplicated().sum() # duplicate rows?
df.describe()     # READ THE MIN AND MAX FIRST
df[col].value_counts() # 'Male', 'male', 'M' — three categories?
df.hist()         # every distribution at once
df.corr()         # linear relationships only
sns.pairplot(df, hue=...) # everything against everything
```



Until next week

1

01

Run and learn our 3 notebooks.

2

02

Investigate your own data.

3

03

Record your findings, you need them for your presentation week.

Read more

- **If you read one thing:**

Wilke, C. O. (2019). Fundamentals of Data Visualization. O'Reilly. Free online, heavily illustrated, and organised around the question-first logic we used today.

- **Healy, K. (2018).**

Data Visualization: A Practical Introduction. Princeton UP. Free online.

- **Cairo, A. (2019).**

How Charts Lie. W. W. Norton.

- **Tukey, J. W. (1977).**

Exploratory Data Analysis. Addison-Wesley. The original.